Frozen Gap Areas and the Sharpness of Source-Cover Certificates

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Abstract

We continue the source-cover analysis of linear-time transposition of a $d \times d$ matrix with $b(d) = \max(1, \lceil \log_2 d \rceil)$-bit entries on a multitape Turing machine. For a fixed admissible return geometry, the source-cover paper of this series bounds the correctness deficit below by the mean circular gap area of the complete raw-source banks, frozen at the first output time of each column, minus an explicit revelation cost. We use this frozen gap area itself as the certificate. Along every joint profile of every near-full sequence, its limiting mean over legal phases and columns is at most $\beta\tau_R/(M-R)$, where τ_R is the limiting hop tail after the first R core families (Theorem G). This implies the source-cover paper's necessary reciprocal-cover bound. We apply Theorem G to two classes of geometries, with no envelope on the sizes of the core families. A *funnel* of depth m is a chain of time sets in which each set is fed by at most one family, and only from the next set, while the last set is fed by none. In no near-full sequence do infinitely many geometries have funnels of a fixed depth that contain every read of a raw interval of fixed positive relative length, whatever the fixed number of exceptional families (Theorem A). A *single-sweep terminal-landing* geometry has two designated families, an older and a newer one, and its exceptional families are among them. It reads the input in one nondecreasing sweep during the lower region of the older one, keeps the input head at one coordinate during the lower regions of the core families and of the newer one and at every column root, and lets none of these families act during the sweep. No near-full sequence consists of such geometries (Theorem B). Theorem G cannot exclude the boundary case of the source-cover paper's power envelopes for two exceptional families. An explicit sequence of admissible geometries with two exceptional families and hop counts at most $512L/i^2$ has mean frozen gap areas at most $10^7/R^2$ at every modulus and hop tails at least $1/(32R)$, so the necessary condition of Theorem G holds on it whenever $3.2 \cdot 10^8(M-R) \le \beta R$, for instance at $M = 2R+2$ for every $\beta \ge 10^9$ (Theorem Z). These geometries carry a single input: they are not near-full, and they say what the certificate can see, not that a transposer exists. All numbered theorems, lemmas, corollaries and definitions are formalized in Lean; the comparison with the reciprocal-cover bound is a remark with a proof sketch only, and a few elementary consequences stated in prose are labelled as not formalized. These are certificates for specified geometries, not a decision of the transposition question.

Contributions. Hans Brende chose the problem and directed the research. Claude Opus 5.5 (Anthropic) developed the mathematics, the Lean formalization and the text. The memo extends and reuses the Lean library of the source-cover paper by Hans Brende and GPT-6 Astra (OpenAI). All numbered theorems, lemmas, corollaries and definitions are formalized in Lean; Remark 3.3 is a proof sketch, and the outlines of the formal proofs are explanatory guides.

Contents

1 Introduction **2**

2 Notation from the source-cover paper **4**

3 Theorem G: the necessary frozen gap area **5**

4 Theorem A: funnels **8**

5 Theorem B: single-sweep terminal landings **9**
5.1 Successful core words 9
5.2 A newer family and a terminal older family 10
5.3 Landings at one selection and phase 10
5.4 Single-sweep terminal-landing geometries 11
5.5 Theorem B 13

6 Sharpness for two exceptional families **14**

7 What remains **19**

A Index of Lean declarations **20**

B Building and checking the proofs **22**

1 Introduction

The question. Let $b(d) = \max(1, \lceil \log_2 d \rceil)$ and $L = d^2 b(d)$. Question (T) asks whether one deterministic multitape Turing machine, with a fixed finite number of tapes and a fixed finite alphabet, transposes every $d \times d$ matrix with $b(d)$-bit entries, given and returned in row-major order, in $O(L)$ steps.

The source-cover paper of this series [2] fixes one complete computation geometry and a nonempty set $\mathcal{C}$ of inputs that it transposes correctly. It bounds the correctness deficit $\Delta = L - \log_2 |\mathcal{C}|$ below by a geometric credit minus the information revealed by a certificate [2, Theorem 3.5]. At each legal phase ρ and column j the credit is at most the circular point-gap area $\mathcal{A}(S_\rho(s_j))$ of the complete raw-source set of an exact source bank, frozen at the first output time s_j of the column [2, Theorems 2.7 and 2.9]. That paper bounds the frozen area below by fractional covers, anchored approximation and shared support. It obtains a uniform exclusion for one exceptional family, and further exclusions under economical-cover or shared-support conditions. For two exceptional families it identifies the inverse-square hop profiles as the nearest unresolved case [2, Section 9.1].

Results.

1. *Theorem G (Theorem 3.2).* We keep the frozen point area itself as the certificate. Along every joint profile of every near-full sequence, and for all integers $R \geq 0$ and $M \geq \max(2, R+1)$,

$$\mathcal{A}^*_{R,M} \leq \frac{\beta \tau_R}{M - R},$$

where $\mathcal{A}^*_{R,M}$ is the limiting mean frozen point area over legal phases and columns, with the first R core families protected (Definition 3.1). The limits are those of the reciprocal-cover bound [2, Theorem 4.6], taken in the same order. Since the frozen point area dominates the fractional-cover credit [2, Lemma 3.3], Theorem G implies that bound (Remark 3.3).

2. *Theorem A (Theorem 4.3).* A funnel is a property of the full family system: time sets $D_0, \ldots, D_m$ such that each D_i with $i < m$ is fed by at most one family, which moves into D_i only times of D_{i+1}, while no family feeds D_m (Definition 4.1). Every orbit then meets D_0 in at most $m + 1$ times (Lemma 4.2). With Theorem G, no near-full sequence has, from some

index on or along a subsequence, funnels of a fixed depth whose first set contains every time that reads a raw interval of fixed positive relative length. The number of exceptional families and the sizes of the core families are arbitrary.

3. *Theorem B (Theorem 5.8).* A single-sweep terminal-landing geometry (Definition 5.5) has two designated families, an older family e and a newer family g, and its exceptional families are among them. Its raw path is constant on the lower regions of all core families and of g and at all column roots, and nondecreasing on the lower region of e; no core family and not g acts on that lower region. No near-full sequence consists, from some index on or along a subsequence, of such geometries. No size envelope is assumed.

4. *Theorem Z (Theorem 6.6 and Corollary 6.7).* At moduli comparable to R, Theorem G cannot exclude the two-exception class with an inverse-square hop envelope $|G_i| \le HL/i^2$, the boundary case of the power envelopes excluded in [2, Corollary 4.4]. We construct admissible geometries Γ_ν with two exceptional families and hop counts $|G_i| \le 512L/i^2$ (Definition 6.1). In every hop order, at every protected prefix $J \ge R$ and every modulus $M > R$, their mean frozen point areas over legal phases and columns are at most $10^7/R^2$, and their hop tails after R are at least $1/(32R)$. Hence $\mathcal{A}^*_{R,M} \le \beta\tau_R/(M-R)$ along every joint profile whenever $3.2 \cdot 10^8(M-R) \le \beta R$, for instance at $M = 2R+2$ for every $\beta \ge 10^9$. The geometries carry one input; frozen point areas and hop profiles do not involve the correctness sets. Two exceptional families thus suffice to meet the frozen gap-area condition at these moduli.

How to read the memo and its formal proofs. Every numbered theorem, lemma and definition is formalized in Lean. Each carries a status line naming its Lean file and declarations, and each theorem and lemma is followed by an outline of the formal proof. The Lean statements, including their inherited hypotheses, are authoritative; the outlines are guides. Remark 3.3, the comparison with the reciprocal-cover bound, is not formalized: it is a remark with a proof sketch only. A few elementary consequences stated in prose, after Theorems G and A and in Sections 5 and 6, are not formalized either, and are labelled so where they occur. Results of the source-cover paper are cited by their numbers at the pinned revision; all of them are Lean-verified there. The memo's library imports that paper's library unchanged (Appendix B).

Related work. The model, the exact source bank, cyclic freezing, the paid inequality, joint profiles and the necessary hop-tail and reciprocal-cover bounds are those of [2]. The finite form of Theorem G is the case of the formal paid inequality of [2, Theorem 3.5], `SourceCovers.paid_deficit_of_gap_credit`, whose credit is the frozen point area itself. The new content of Theorem G is its limit form and its use in Sections 4 and 5. The counting there rests on the orbit structure of [2, Lemma 2.6] and on the interval bound recalled in Section 2. We claim no novelty for these elementary covering and counting principles.

Organization. Section 2 recalls the notation. Section 3 proves Theorem G, Section 4 Theorem A and Section 5 Theorem B. Section 6 constructs the sharpness geometries and proves Theorem Z, and Section 7 states what remains. The appendices index the Lean declarations and give the build and check commands.

2 Notation from the source-cover paper

We use the notation of [2] and recall only what the statements below need. All objects in this section are defined and Lean-verified there.

Geometries. Fix integers $d, b \geq 1$ and put $L = d^2 b$, $W = db$ and $\Omega = \{0, \dots, L-1\}$; for (T), $b = b(d)$. A fixed complete geometry consists of the input-head and work-head walks, the halting time T and the append times $t_0 < \cdots < t_{L-1} < T$ [2, Section 2.1]. The *raw path* x is the input-head position clamped to Ω; it moves by at most one cell per step. In the Lean model times are natural numbers, and statements about all times quantify over all of them. The first append time of column j is $s_j = t_{jW}$ [2, Theorem 2.9]. An *admissible geometry* with parameters $d, b, c, \beta, N, q, a_0$ [2, Definition 2.2] adds a nonempty correctness set $\mathcal{C}$, the runtime bound $T \leq cL$, whole-step record capacity at most β, an N-class span partition of the full family system, an original Q_2 core $\mathcal{K}_0$, a disjoint set $\mathcal{E}$ of at most q exceptional complete families, and the bound a_0 on the hop count of every other family. Its correctness deficit is $\Delta = L - \log_2 |\mathcal{C}|$ [2, (2.2)].

Families and shadows. A complete actual family F has an owner o_F, an outermost hop (p_F, U_F), a full span $(p_F, U_F]$ and a hop count $|F|$ [2, Definition 2.1]. We call $(p_F, o_F]$ its *lower region* and $(o_F, U_F]$ its *active interval*. Its clipped shadow s_F is the identity off the active interval and moves every time of it; by [2, Definition 2.4 and (2.6)],

$$p_F < s_F(t) \leq o_F < t \leq U_F \qquad \text{whenever } o_F < t \leq U_F; \tag{2.1}$$

on its active interval s_F is antitone. We write $F(t) = s_F(t)$ and say that F *acts* at t if $F(t) \neq t$, that is, if t lies in its active interval. For a set $\mathcal{S}$ of families, $\mathcal{O}_{\mathcal{S}}(t)$ is the set of values obtained from t by finite compositions of their shadows, including the empty composition [2, Definition 2.4]; its elements are at most t. A word $(F_1, \dots, F_n)$ of families is *successful* from t if F_1 acts at t, F_2 acts at $F_1(t)$, and so on; its *endpoint* is $F_n(\cdots F_1(t) \cdots)$.

Cores, phases and banks. A set $\mathcal{K}$ of complete families is Q_2 if its spans split into two classes of pairwise disjoint spans. Joining two families of $\mathcal{K}$ when their spans intersect gives a caterpillar forest, the *overlap forest*; its *original nonleaves* are the vertices with at least two neighbours. The nonleaves of each component form a path, so a nonleaf has at most two nonleaf neighbours, and a leaf has at most one neighbour [2, Lemma 2.6]. Fix $M \geq 2$ and a protected set of at most $R < M$ families of $\mathcal{K}$. For a residue $\rho \in \{0, \dots, M-1\}$, the original nonleaves whose rank along their nonleaf path is congruent to ρ modulo M are deleted, and the other families of $\mathcal{K}$ *survive*; they form the retained core. A residue is a *legal phase* if no protected nonleaf has a rank congruent to it; there are at least $M - R$ legal phases. Let $A_\rho(u)$ be the orbit of the retained core at u. Every successful word of surviving families has at most $M + 1$ letters, and $|A_\rho(u)| \leq 2M + 3$ [2, Lemma 2.6]. At most two families of $\mathcal{K}$ act at any one time, since their spans contain that time and so lie in distinct classes. For an exceptional set $\mathcal{E}$, the exact source bank has value set $O_\rho(t)$ at t, the orbit of the retained core together with the exceptional families [2, Theorem 2.7]. The raw source set is $S_\rho(t) = x(O_\rho(t))$ [2, Section 3.1]; it contains $x(t)$. In limit statements, $\mathcal{K}$ consists of the first J families of the original core in its supplied order by nonincreasing hop count, the protected set consists of the first R, and the forest, ranks and phases are rebuilt for every selection [2, Section 4.2].

Gap area and the paid error. View Ω as the integer points of a circle of circumference L. For a nonempty $S \subseteq \Omega$ whose consecutive distinct points have circular gaps $g_1, \dots, g_v$, the point-gap

area is [2, (2.14)]

$$\mathcal{A}(S) = \frac{1}{2L^2}\sum_{h=1}^{v} g_h^2 \in \Big[0, \frac{1}{2}\Big].$$

We call $\mathcal{A}(S_\rho(s_j))$ the *frozen point area* of phase ρ and column j. One counting fact is used repeatedly. If a nonempty $S \subseteq \Omega$ lies in the union of at most n closed integer intervals $[a_i, b_i]$ with $\sum_i (b_i - a_i) \le \alpha L$, then

$$\mathcal{A}(S) \ge \frac{(1-\alpha)_+^2}{2n}, \qquad (z)_+ = \max(z, 0). \tag{2.2}$$

This is the exposed-gap step in the proof of [2, Lemma 3.3]: an interval contains the left endpoint of at most one gap that it does not contain, the exposed gaps therefore have total length at least $(1-\alpha)_+ L$, and the Cauchy–Schwarz inequality applies to them. Its Lean form is `LeanMachine.IntervalGapArea.point_area_lower_of_budgets` in the shared library. The paid error of [2, (3.7)] is

$$\mathsf{E}_q(w; J, M, R, a, \tau) = \frac{2C_q(T-1)}{dL} + \frac{2(w-1)}{L} + \frac{W-b+1}{2L} + \frac{\beta}{L}\Big[J + q + \frac{\tau}{M-R} + \frac{(1+Na)C_q(T-1)}{w}\Big],$$

with $C_q = C_0(1+3C_0)^q$ and $C_0 = 16 + 24M(M-1)$.

Near-full sequences and joint profiles. A near-full sequence consists of admissible geometries indexed by ν, with common c, β, N, q, such that $d_\nu \to \infty$, $a_{0,\nu}/L_\nu \to 0$ and $\Delta_\nu/L_\nu \to 0$ [2, Definition 4.1]. Each geometry comes with an order $G_1, G_2, \dots$ of its original core by nonincreasing hop count. A joint profile is a coordinatewise limit $(h_i, v_i)_{i\ge1}$ of the normalized profiles $h_i^{(\nu)} = |G_i|/L_\nu$ and $v_i^{(\nu)} = \mathrm{Var}(x_\nu; [p_{G_i}, U_{G_i}])/L_\nu$ along one strictly increasing subsequence; both profiles are zero after the last family. Write

$$\tau_R = \sum_{i>R} h_i, \qquad \tau_{R,J} = \sum_{R<i\le J} h_i, \qquad V_K = \sum_{i>K} v_i.$$

The limiting profiles are summable, so $\tau_R \to 0$ as $R \to \infty$. Limits in ν are taken along the subsequence of the profile. A strictly increasing subsequence of a near-full sequence is near-full, and every near-full sequence has a joint profile [2, Section 4.1].

3 Theorem G: the necessary frozen gap area

The paid inequality of [2, Theorem 3.5] accepts any credit that is at most the frozen point area at each legal phase and column. Here the credit is the frozen point area itself, and the limits are taken as for the reciprocal-cover bound [2, Theorem 4.6].

Definition 3.1 (Frozen gap-area means)**.** Lean: `SourceCovers.phaseGapArea`, `SourceCovers.phaseGapArea_bounds`, `SourceCovers.meanGapArea`, `SourceCovers.AdmissibleGeometry.prefixMeanGapArea`, `SourceCovers.NearFullSequence.JointProfile.prefixGapArea`, `SourceCovers.NearFullSequence.JointProfile.gapAreaLimit`, `SourceCovers.NearFullSequence.JointProfile.gapAreaLimit_bounds` [`GapAreaMemo/GapArea.lean`]

Fix a geometry, a Q_2 selected core, an exceptional set, $M \ge 2$ and a protected set of at most $R < M$ selected families. For a residue $\rho < M$ and a column $j < d$, the frozen point area $\mathcal{A}(S_\rho(s_j))$ lies in $[0, 1/2]$. Let $\overline{\mathcal{A}}$ be its arithmetic mean over the legal phases ρ and all columns j. For an admissible

geometry and an order of its original core by nonincreasing hop count, $\overline{\mathcal{A}}_{J,R,M}$ denotes this mean when the first J original core families are selected and the first R protected. Along a joint profile of a near-full sequence, $\overline{\mathcal{A}}_{\nu,J,R,M}$ is this mean for the geometry of index ν on the subsequence of the profile, and

$$\mathcal{A}^*_{R,M} = \limsup_{J\to\infty} \limsup_{\nu\to\infty} \overline{\mathcal{A}}_{\nu,J,R,M}, \qquad \mathcal{A}^*_R = \mathcal{A}^*_{R,2R+2}. \tag{3.1}$$

For $R < M$ both upper limits are real, and $0 \le \mathcal{A}^*_{R,M} \le 1/2$.

Theorem 3.2 (Theorem G: necessary frozen gap area)**.** `Lean`:
`SourceCovers.gap_area_paid_deficit`,
`SourceCovers.AdmissibleGeometry.prefix_gap_area_deficit`,
`SourceCovers.NearFullSequence.JointProfile.prefix_gap_area_limit`,
`SourceCovers.NearFullSequence.JointProfile.necessary_gap_area`,
`SourceCovers.NearFullSequence.JointProfile.necessary_gap_area_default`
[`GapAreaMemo/GapArea.lean`]

(a) Finite form. *Take the data of [2, Theorem 3.5]: a fixed complete geometry with a nonempty correctness set, whole-step capacity at most β and an N-class span partition of its full family system; a Q_2 selected core of at most J families; at most q exceptional families; $M \ge 2$ and a protected set of at most $R < M$ selected families; a bound a on the hop count of every family outside the selected core and the exceptions; and a real τ at least the hop mass of the unprotected original nonleaves of the selected core. Then, for every integer $w \ge 1$,*

$$\frac{\Delta}{L} \ge \overline{\mathcal{A}} - \mathsf{E}_q(w; J, M, R, a, \tau). \tag{3.2}$$

No runtime bound is used.

(b) Prefixes. *For an admissible geometry with any order $G_1, G_2, \dots$ of its original core by nonincreasing hop count, and integers $J \ge R \ge 0$, $M \ge \max(2, R+1)$, $w \ge 1$ and $a \ge a_0$ with $a \ge cL/(J+1)$,*

$$\frac{\Delta}{L} \ge \overline{\mathcal{A}}_{J,R,M} - \mathsf{E}_q\Big(w; J, M, R, a, \sum_{R<i\le J} |G_i|\Big).$$

(c) Dimension limit. *Along every joint profile of every near-full sequence, for all integers $J \ge R \ge 0$ and $M \ge \max(2, R+1)$ and every real $\lambda > 0$,*

$$\limsup_{\nu\to\infty} \overline{\mathcal{A}}_{\nu,J,R,M} \le \frac{\beta\tau_{R,J}}{M-R} + 2\lambda + \frac{\beta N C_q c^2}{\lambda(J+1)}.$$

(d) The limit bound. *For every joint profile of every near-full sequence and all integers $R \ge 0$ and $M \ge \max(2, R+1)$,*

$$\mathcal{A}^*_{R,M} \le \frac{\beta\tau_R}{M-R}. \tag{GA}$$

In particular $\mathcal{A}^*_R \le \beta\tau_R/(R+2)$.

In (c) and (d) the cutoff R and the modulus M are fixed, and the number q of exceptional families is any fixed integer.

Outline of the formal proof. (a) Apply `SourceCovers.paid_deficit_of_gap_credit`, the formal paid inequality of [2, Theorem 3.5], with the frozen point area $\mathcal{A}(S_\rho(s_j))$ as the credit of each legal

phase ρ and column j. Its premise, that each credit is at most the frozen point area, holds with equality. The mean credit is then $\overline{\mathcal{A}}$, and the paid error is unchanged.

(b) The first J original families form a Q_2 selection of at most J families, and the hop mass of its unprotected nonleaves is at most $\sum_{R<i\le J}|G_i|$. An omitted core family has at most $cL/(J+1)$ hops, by the sorted order and the total bound $\sum_i |G_i| \le cL$; every other non-exceptional family has at most a_0 hops. Hence (a) applies with this a, as in the proof of [2, Theorem 4.2].

(c) For fixed J, R, M, λ, the error-limit lemma of the source-cover paper, `SourceCovers.NearFullSequence.JointProfile.prefix_error_limit`, supplies along the subsequence widths $w_\nu \ge 1$ and values a_ν admissible in (b), with widths $\lceil \lambda L_\nu \rceil$ as in the proof of [2, Theorem 4.2], and upper bounds F_ν for the paid errors such that $\Delta_\nu/L_\nu + F_\nu$ tends to the right side of (c). By (b), $\overline{\mathcal{A}}_{\nu,J,R,M} \le \Delta_\nu/L_\nu + F_\nu$, and the means lie in $[0, 1/2]$; pass to the upper limit.

(d) Let $J \to \infty$ in (c): $\tau_{R,J} \to \tau_R$ by summability, and the last term tends to zero. The finite upper limits lie in $[0, 1/2]$, so the outer upper limit preserves the inequality. Finally take $\lambda = \delta/2$ for every $\delta > 0$ and remove the additive δ. For $M = 2R+2$, $M - R = R + 2$. As in [2, Theorem 4.6], the dimension limit, the prefix limit and the removal of the width error occur in that order. □

Why it matters. Theorem G is the necessary condition that the paid inequality gives without choosing a cover witness. For an impossibility proof it reduces the exclusion of a class of admissible geometries to one lower bound: it suffices to show that every near-full sequence in the class has a joint profile with $\mathcal{A}^*_{R,M} > \beta\tau_R/(M-R)$ for some R and M. Such a bound concerns the geometries alone, since frozen point areas and hop profiles do not involve the correctness sets. Theorems A and B prove such bounds for two classes. For a construction it is a requirement on the frozen banks. For example, $\mathcal{A}(S) \ge 1/(2|S|)$ for every nonempty S by the Cauchy–Schwarz inequality. Along a joint profile of a near-full sequence with an inverse-square limiting profile $h_i \le H/i^2$, for which $\tau_R \le H/R$ when $R \ge 1$, the reciprocal sizes $1/|S_\rho(s_j)|$ of the frozen raw sets therefore have limiting mean at most $2\beta H/(R(R+2))$ at modulus $2R+2$ for every $R \ge 1$, in the sense of (3.1). This consequence is elementary and is not formalized.

Remark 3.3 (Comparison with the reciprocal-cover bound)**.** Proof sketch only: for every joint profile of every near-full sequence, and more generally of every sequence of admissible geometries with $d \to \infty$ (the argument does not use near-fullness, and $Q^*_{R,M}$ is defined along the profile as in [2, (4.7)]), and all integers $K, R \ge 0$ and $M \ge \max(2, R+1)$,

$$\frac{(1-V_K)_+^2}{2(2K+1)}\, Q^*_{R,M} \le \mathcal{A}^*_{R,M},$$

where $Q^*_{R,M}$ is the reciprocal-cover limit of [2, (4.7)]. Hence (GA) implies the weighted reciprocal-cover bound [2, (4.8)], and through it the necessary hop-tail bound [2, Theorem 4.2]. Indeed, for $J \ge \max(K, R)$ distinguish the first K selected families. At every legal phase and column, [2, Lemma 3.3] gives

$$\mathcal{A}(S_\rho(s_j)) \ge \frac{(1-\alpha_\nu)_+^2}{2(2K+1)\kappa_{\nu,J,R,M,\rho,j}}, \qquad \alpha_\nu = \sum_{K<i\le J} v_i^{(\nu)},$$

with the fractional cover κ of [2, Definition 3.1]; its formal form, `SourceCovers.fractional_phase_gap`, is stated for the same point area. Average over legal phases and columns. Along the profile $\alpha_\nu \to \sum_{K<i\le J} v_i \le V_K$, and all averages lie in $[0,1]$, so the dimension upper limit of $\overline{\mathcal{A}}_{\nu,J,R,M}$ is at least $(1-V_K)_+^2/[2(2K+1)]$ times that of the mean reciprocal cover. The prefix upper limit preserves this. Theorem G retains the full frozen point area, including what the fractional-cover estimate gives up through the factors $(1-\alpha)_+^2$ and $2(2K+1)$.

4 Theorem A: funnels

A funnel restricts how shadows of the full family system can enter the times at which a raw interval is read. It involves no root, selection or phase.

Definition 4.1 (Feeding and funnels)**.** Lean: `SourceCovers.ActualFamily.Feeds`, `SourceCovers.Funnel`, `SourceCovers.AdmissibleGeometry.HasFunnel` [`GapAreaMemo/Funnels.lean`]

Fix a geometry. A complete family F *feeds* a set D of times if $F(t) \neq t$ and $F(t) \in D$ for some time t. For an integer $m \geq 0$, a *funnel of depth* m consists of sets $D_0, \dots, D_m$ of times and complete families $f_0, \dots, f_{m-1}$ such that

(F1) for each $i < m$, every complete family that feeds D_i is f_i;

(F2) for each $i < m$ and every time t with $f_i(t) \neq t$ and $f_i(t) \in D_i$, the time t lies in D_{i+1};

(F3) no complete family feeds D_m.

Here the complete families range over the whole family system: core, exceptional and all other families. The formal definition indexes the sets and families by all $i \geq 0$, of which only $D_0, \dots, D_m$ and $f_0, \dots, f_{m-1}$ matter; it therefore presupposes a nonempty family system. An admissible geometry *has a funnel of depth* m *over a raw interval of relative length* δ if there are such a funnel and integers $0 \leq z_0 \leq z_1 \leq L$ with $z_1 - z_0 \geq \delta L$ such that every time s with $z_0 < x(s) < z_1$ lies in D_0.

For $m = 0$, conditions (F1) and (F2) are vacuous: a funnel of depth zero is a set D_0 of times into which no shadow moves any time, and every orbit meets it at most in its root. For larger m, the set D_i can be entered only through f_i and only from D_{i+1}, and the chain ends in a set that nothing enters.

Lemma 4.2 (Lemma A: a funnel forces point area)**.** Lean: `SourceCovers.Funnel.reach`, `SourceCovers.Funnel.reach_card`, `SourceCovers.Funnel.orbit_inter_subset`, `SourceCovers.phaseGapArea_ge_of_funnel`, `SourceCovers.AdmissibleGeometry.prefixMeanGapArea_ge_of_funnel` [`GapAreaMemo/Funnels.lean`]

Fix a geometry with a funnel of depth m.

(a) For every set $\mathcal{S}$ *of complete families and every time* s*, the orbit* $\mathcal{O}_{\mathcal{S}}(s)$ *meets* D_0 *in at most* $m+1$ *times. More precisely,* $\mathcal{O}_{\mathcal{S}}(s) \cap D_0 \subseteq P_m(s)$*, where* $P_0(s) = \{s\}$ *and* $P_{n+1}(s) = \{s\} \cup f_{m-n-1}(P_n(s))$ *for* $n < m$*, so that* $|P_n(s)| \leq n+1$.

(b) Let $0 \leq z_0 \leq z_1 \leq L$ *be integers such that every time* s *with* $z_0 < x(s) < z_1$ *lies in* D_0*. Then for every* Q_2 *selected core, every exceptional set, every* $M \geq 2$*, every residue* $\rho < M$ *and every column* $j < d$*,*

$$\mathcal{A}(S_\rho(s_j)) \geq \frac{\big((z_1 - z_0)/L\big)^2}{2(m+3)}.$$

(c) If an admissible geometry has a funnel of depth m *over a raw interval of relative length* $\delta \geq 0$*, then* $\overline{\mathcal{A}}_{J,R,M} \geq \delta^2/(2(m+3))$ *for every hop order and all* J*,* R *and* $M \geq \max(2, R+1)$.

Outline of the formal proof. (a) We show that every word over families of $\mathcal{S}$ whose endpoint from s lies in D_{m-n} has its endpoint in $P_n(s)$, by induction on the word from its last letter. The empty word ends at $s \in P_n(s)$, and a last letter that does not act leaves the endpoint unchanged. Suppose the last letter F moves the previous endpoint u into D_{m-n}. Then F feeds D_{m-n}, which (F3) excludes if $n = 0$. If $n \geq 1$, then $F = f_{m-n}$ by (F1), and $u \in D_{m-n+1}$ by (F2). By induction $u \in P_{n-1}(s)$, so the endpoint lies in $f_{m-n}(P_{n-1}(s)) \subseteq P_n(s)$. Take $n = m$.

(b) By the exactness of the bank [2, Theorem 2.7], every raw value of $S_\rho(s_j)$ strictly between z_0 and z_1 is the value of x at an orbit point in D_0; by (a) there are at most $m+1$ such values. Hence $S_\rho(s_j)$ lies in $[0, z_0]$, $[z_1, L]$ and at most $m+1$ singletons: at most $m+3$ closed intervals of total length $L-(z_1-z_0)$. Apply (2.2) with $\alpha = 1-(z_1-z_0)/L$.

(c) The bound of (b) holds at every legal phase and column; there is at least one legal phase. Average. □

Theorem 4.3 (Theorem A: funnels are excluded)**.** Lean: `SourceCovers.NearFullSequence.not_eventually_funnel`, `SourceCovers.NearFullSequence.not_frequently_funnel` [`GapAreaMemo/Funnels.lean`]

Fix an integer $m \geq 0$ and a real $\delta > 0$. In no near-full sequence do all geometries from some index on have a funnel of depth m over a raw interval of relative length δ, and in none do infinitely many of its geometries have one. Funnels are as in Definition 4.1; in particular the family system is nonempty. The parameters c, β, N, q of the sequence are arbitrary, and no envelope on the sizes of the core families is assumed.

Outline of the formal proof. Choose a joint profile and put $B = \delta^2/(2(m+3)) > 0$. If the geometries have such funnels from some index on, then so do those on the subsequence of the profile, and Lemma 4.2(c) with $R = 0$ gives $\mathcal{A}^*_{0,M} \geq B$ for every $M \geq 2$. Theorem G with $R = 0$ gives $\mathcal{A}^*_{0,M} \leq \beta\tau_0/M$. An $M \geq 2$ with $BM > \beta\tau_0$ gives a contradiction. If funnels occur in infinitely many geometries, these geometries form a strictly increasing subsequence, which is again a near-full sequence, and the first case applies to it. □

Why it matters. Theorem A excludes a property of the full family system with no hypothesis on the number of exceptional families or on the sizes of the core families. For an impossibility proof, it shows that bounded-depth unique feeding of the reads of a macroscopic raw interval already contradicts near-full correctness. For a construction in the admissible class it is a requirement, which can be read off by following feeders backwards. Let X be the set of times that read the open interval (z_0, z_1), put $D_0 = X$, and as long as D_i is fed by exactly one family f_i, let D_{i+1} be the set of times that f_i moves into D_i. If some D_i with $i \leq m$ is fed by no family, then $D_0, \ldots, D_i$, followed by empty sets with arbitrary families, form a funnel of depth m over the interval; this needs a nonempty family system, and an empty one is excluded directly by Theorem G, since then every frozen raw set is a single point, of area $1/2$, while $\tau_R = 0$. So, in a near-full sequence, from some index on and for every raw interval of length at least δL, this backward chain either reaches, among $D_0, \ldots, D_m$, a set fed by at least two families, or has $D_0, \ldots, D_m$ each fed by exactly one family. This reformulation is elementary and is not formalized.

5 Theorem B: single-sweep terminal landings

The class of this section has two designated families: an older family e, whose lower region carries the only raw sweep, and a newer family g. We first count orbit points of a Q_2 selected core, then bound the frozen point area at one selection and phase, and finally combine two selections with Theorem G.

5.1 Successful core words

Lemma 5.1 (Successful core words)**.** Lean: `SourceCovers.coreSuccess_prefix_or`, `SourceCovers.coreSuccess_internal_nonleaf`, `SourceCovers.core_root_count` [`GapAreaMemo/CoreWords.lean`]

Let $\mathcal{K}$ be a Q_2 set of complete families, and consider words of families of $\mathcal{K}$.

(a) Two words that are successful from the same time and have the same first letter are comparable: one is a prefix of the other.

(b) Every letter of a successful word other than its first and its last letter is an original nonleaf of the overlap forest of $\mathcal{K}$.

(c) Let r be a time, X a finite set of times and $\Phi \subseteq \mathcal{K}$. Suppose that each $v \in X$ is the endpoint of a nonempty word w_v that is successful from r, and that the last letter of w_v belongs to Φ whenever it is a nonleaf. Then $|X| \le 2|\Phi| + 4$.

Part (b) holds for every set of complete families.

Outline of the formal proof. (a) Induct on the words. After a nonidentity step of a family, the new time lies in its span at or before its owner. That family no longer acts there, no other family of its class acts there, and at most one family of the other class does [2, proof of Lemma 2.6]; the formal statement is `SourceCovers.core_unique_after`. So the second letters agree whenever both words continue.

(b) Consecutive letters of a successful word have intersecting spans and strictly decreasing owners, so the word is a simple path of the overlap forest [2, Lemma 2.6]. An internal letter has two distinct neighbours on that path.

(c) Consider first the points whose words end in a nonleaf. Two such words with the same first and last letters are comparable by (a), and the letters of a successful word are distinct, so the words are equal. The first letter acts at r, and at most two families act there; this gives at most $2|\Phi|$ points. For words that end in a leaf, two words of length at least two with the same first letter are comparable by (a), and a proper prefix would make its last letter, a leaf, an internal letter of the longer word, contrary to (b). Recording the first letter and whether the length is one gives at most four such points. □

5.2 A newer family and a terminal older family

Lemma 5.2 (Adding a newer and a terminal older family)**.** Lean: `SourceCovers.shadowOrbit_insert_newer`, `SourceCovers.shadowOrbit_insert_terminal`, `SourceCovers.raw_landing_antitone` [`GapAreaMemo/LandingOrbits.lean`]

Let $\mathcal{F}$ be a set of complete families, g and e complete families, and s a time.

(a) Every point of $\mathcal{O}_{\mathcal{F}\cup\{g\}}(s)$ lies in $\mathcal{O}_{\mathcal{F}}(s)$, or in $\mathcal{O}_{\mathcal{F}}(g(u))$ for some $u \in \mathcal{O}_{\mathcal{F}}(s)$ at which g acts.

(b) Suppose that no family of $\mathcal{F}$ acts at any time of the lower region $(p_e, o_e]$. Then every point of $\mathcal{O}_{\mathcal{F}\cup\{e\}}(s)$ lies in $\mathcal{O}_{\mathcal{F}}(s)$ or equals $e(u)$ for some $u \in \mathcal{O}_{\mathcal{F}}(s)$ at which e acts.

(c) If x is nondecreasing on $(p_e, o_e]$, then $x(e(u')) \le x(e(u))$ whenever $o_e < u \le u' \le U_e$.

Outline of the formal proof. Parts (a) and (b) follow by induction over the letters of a word. (a) After a nonidentity step of g the time is at most o_g, and no later step increases it, so g never acts again. (b) After a nonidentity step of e the time lies in $(p_e, o_e]$ by (2.1), where neither e nor a family of $\mathcal{F}$ acts; the point is final. (c) On its active interval the shadow of e is antitone with values in $(p_e, o_e]$; compose with the nondecreasing raw path. □

5.3 Landings at one selection and phase

Definition 5.3 (Newer closure, landings and starts)**.** Lean: `SourceCovers.newerRoots`, `SourceCovers.newerClosure`, `SourceCovers.landingSet`, `SourceCovers.IsLandingStep`,

`SourceCovers.stepJump`, `SourceCovers.stepMax`,
`SourceCovers.landingStarts` [`GapAreaMemo/LandingCounts.lean`]

Fix a Q_2 selected core $\mathcal{K}$, $M \geq 2$, a residue $\rho < M$, complete families g and e, and a time s. The *newer roots* of s are $N_\rho(s) = \{s\} \cup \{g(u) : u \in A_\rho(s),\ g(u) \neq u\}$, and its *newer closure* is $C_\rho(s) = \bigcup_{r \in N_\rho(s)} A_\rho(r)$. Since orbits are closed under composition, this is the set $C_g(s)$ of [2, Proposition 3.7(b)], with g in the role of the newer exceptional map. Its *landing times* are $\Lambda_\rho(s) = C_\rho(s) \cap (o_e, U_e]$. A *landing step* is a pair (u, F) of a landing time u and a surviving family F that acts at u with $F(u) \in (o_e, U_e]$. Its *jump* is $x(e(F(u))) - x(e(u))$, truncated at zero, and $\Gamma_\rho(s)$ is the largest jump of a landing step, or zero if there is none. The *starts* $\Lambda^0_\rho(s)$ are the landing times v for which there is no landing time u and surviving family F acting at u with $F(u) = v$.

Lemma 5.4 (Counts at one selection and phase)**.** Lean: `SourceCovers.newerClosure_count`, `SourceCovers.newerClosure_count_span`, `SourceCovers.landingStarts_card`, `SourceCovers.landing_cover` [`GapAreaMemo/LandingCounts.lean`]

In the setting of Definition 5.3:

(a) If D is a set of times and $\Phi \subseteq \mathcal{K}$ contains every original nonleaf whose lower region meets D, then $|C_\rho(s) \cap D| \leq (2M + 4)(2|\Phi| + 5)$.

(b) If $F \in \mathcal{K}$, $p_F < a$ and $v \leq U_F$, then $|C_\rho(s) \cap (a, v)| \leq 11(2M + 4)$.

(c) $|\Lambda^0_\rho(s)| \leq 5(2M + 4)$.

(d) If x is nondecreasing on $(p_e, o_e]$, then every $v \in \Lambda_\rho(s)$ has a start $s_0 \in \Lambda^0_\rho(s)$ with

$$x(e(s_0)) \leq x(e(v)) \leq x(e(s_0)) + (M + 1)\,\Gamma_\rho(s).$$

Outline of the formal proof. (a) There are at most $1 + |A_\rho(s)| \leq 2M + 4$ newer roots. Fix one root r. Erasing inactive letters [2, Lemma 2.6], every point of $A_\rho(r)$ other than r is the endpoint of a nonempty successful word of surviving families from r. By (2.1) the endpoint lies in the lower region of the last letter, so for an endpoint in D a nonleaf last letter belongs to Φ. Lemma 5.1(c) leaves at most $2|\Phi| + 4$ such points, and with r at most $2|\Phi| + 5$ points per root.

(b) Take for Φ the family F together with the nonleaves adjacent to F. A nonleaf whose lower region meets $(a, v) \subseteq (p_F, U_F]$ is F or has a span meeting that of F. A nonleaf has at most two nonleaf neighbours and a leaf has at most one neighbour, so $|\Phi| \leq 3$, and (a) gives $11(2M + 4)$.

(c) Every start is a newer root, or lies in $A_\rho(r)$ for a newer root r and differs from it. In the second case it is the endpoint of a nonempty successful survivor word from r. Its last step starts at a point of the newer closure that is larger than the start and is not a landing time, since otherwise the start would be reached by a landing step; so that point exceeds U_e. The span of the last letter therefore contains the boundary between U_e and $U_e + 1$, which at most two families of $\mathcal{K}$ do. The first letter acts at r, so it is one of at most two families. Two such words with the same first and last letters are equal, by Lemma 5.1(a) and the distinctness of the letters of a successful word. Each root thus contributes at most $1 + 4$ starts.

(d) Use strong induction on $U_e - v$. A landing time v that is not a start equals $F(u)$ for a landing time $u > v$ and a surviving family F; then (u, F) is a landing step, which raises the raw value by at most $\Gamma_\rho(s)$. The chain of steps from a start s_0 to v is a successful word of surviving families, so it has at most $M + 1$ letters [2, Lemma 2.6]. The lower bound is Lemma 5.2(c), since $v \leq s_0$. □

5.4 Single-sweep terminal-landing geometries

Definition 5.5 (Single-sweep terminal-landing geometry)**.** Lean: `SourceCovers.AdmissibleGeometry.TerminalLanding` [`GapAreaMemo/LandingGaps.lean`]

Single-sweep terminal-landing data for an admissible geometry, with original core $\mathcal{K}_0$ and exceptional set $\mathcal{E}$, consist of complete families e and g, the *older* and the *newer* family, and a natural number x_0, such that

(L1) $\mathcal{E} \subseteq \{e, g\}$;

(L2) $x(t) = x_0$ whenever t lies in the lower region of a family of $\mathcal{K}_0$ or of g;

(L3) $x(s_j) = x_0$ for every column $j < d$;

(L4) x is nondecreasing on the lower region $(p_e, o_e]$ of e;

(L5) no family of $\mathcal{K}_0$, and not g, acts at any time of $(p_e, o_e]$.

A geometry with such data is a *single-sweep terminal-landing geometry*. The families e and g need not be distinct, and need not both be exceptional. Condition (L5) holds for g whenever $o_e \le o_g$ (an elementary remark, not formalized).

Thus the input is read in one nondecreasing sweep during the lower region of e, the input head is parked at x_0 at all other times that can contribute to a frozen bank (Lemma 5.6(a)), and no family of a bank moves a landing of e again.

Lemma 5.6 (Finite gap bounds for terminal landings)**.** `Lean`:
`SourceCovers.AdmissibleGeometry.TerminalLanding.raw_values`,
`SourceCovers.AdmissibleGeometry.TerminalLanding.gap_of_step`,
`SourceCovers.AdmissibleGeometry.TerminalLanding.gap_of_starts`
[`GapAreaMemo/LandingGaps.lean`]

Let an admissible geometry carry single-sweep terminal-landing data (e, g, x_0). Let $\mathcal{K} \subseteq \mathcal{K}_0$ be a Q_2 selected core, $M \ge 2$, $\rho < M$ a residue and $j < d$ a column. Landing sets refer to $\mathcal{K}$, M, ρ, g, e and the root s_j, and $S_\rho(s_j)$ to the geometry's exceptional set.

(a) Every element of $S_\rho(s_j)$ is x_0 or $x(e(v))$ for some $v \in \Lambda_\rho(s_j)$.

(b) Let $F \in \mathcal{K}$ act at a time $u < T$ with $o_e < u \le U_e$ and $o_e < F(u) \le U_e$, and let $\gamma = x(e(F(u))) - x(e(u))$, truncated at zero. Then

$$\mathcal{A}(S_\rho(s_j)) \ge \frac{(\gamma/L)^2}{2(11(2M+4)+3)}.$$

The family F need not survive at ρ.

(c) With $\Gamma = \Gamma_\rho(s_j)$,

$$\mathcal{A}(S_\rho(s_j)) \ge \frac{\left(1 - 5(2M+4)(M+1)\Gamma/L\right)_+^2}{2(5(2M+4)+1)}.$$

Outline of the formal proof. (a) By (L1), the orbit $O_\rho(s_j)$ of the retained core with the exceptions lies in the orbit of the retained core together with g and e. By (L5) and Lemma 5.2(b), each of its points lies in the orbit of the retained core with g, or equals $e(u)$ for a point u of that orbit at which e acts. A point of the first kind is the root, where $x = x_0$ by (L3), or lies in the lower region of a retained family or of g, where $x = x_0$ by (L2). For the second kind, $u \in C_\rho(s_j)$ by Lemma 5.2(a), and $u \in (o_e, U_e]$ because e acts at u.

(b) Since $F(u) < u$, Lemma 5.2(c) gives $x(e(u)) \le x(e(F(u)))$. By (a) and the same monotonicity, every raw value strictly between them is x_0 or $x(e(v))$ for a landing time v with $F(u) < v < u$. Since F acts at u, $p_F < F(u)$ and $u \le U_F$, so Lemma 5.4(b) leaves at most $1 + 11(2M+4)$ raw

values in that open interval. Cover the raw set by the two complementary closed intervals and these singletons, and apply (2.2); the formal step is `SourceCovers.pointArea_ge_of_sparse`.

(c) By (a) and Lemma 5.4(d), the raw set lies in $\{x_0\}$ and the closed intervals $[x(e(s_0)), x(e(s_0)) + (M+1)\Gamma]$ with $s_0 \in \Lambda^0_\rho(s_j)$. By Lemma 5.4(c) these are at most $1 + 5(2M+4)$ intervals of total length at most $5(2M+4)(M+1)\Gamma$. Apply (2.2). □

5.5 Theorem B

Lemma 5.7 (Prefix bounds for terminal landings). Lean:
`SourceCovers.AdmissibleGeometry.landingGamma`,
`SourceCovers.AdmissibleGeometry.landingEnergy`,
`SourceCovers.AdmissibleGeometry.landingEnergy_bounds`,
`SourceCovers.AdmissibleGeometry.prefixMeanGapArea_ge_later`,
`SourceCovers.AdmissibleGeometry.prefixMeanGapArea_ge_own`
[`GapAreaMemo/TerminalLandings.lean`]

Let an admissible geometry carry single-sweep terminal-landing data, and fix an order of its original core by nonincreasing hop count. For $J_0 \geq 0$ and a column j, let $\Gamma_{J_0}(j)$ be the largest of the jumps $\Gamma_\rho(s_j)$ over all selections of the first $J \leq J_0$ original core families with the first $R \leq J$ protected and modulus $2R+2$, and over the legal phases ρ of each selection. The landing energy

$$E(J_0) = \frac{1}{d} \sum_{j<d} \Big(\frac{\Gamma_{J_0}(j)}{L} \Big)^2$$

lies in $[0,1]$. Put $f_R = 20(R+2)(2R+3)$. For every $R \geq 0$:

(a) $\overline{\mathcal{A}}_{J,R,2R+2} \geq E(J_0)/(88R+182)$ for every $J \geq J_0$;

(b) $\overline{\mathcal{A}}_{J,R,2R+2} \geq (1/2 - 2f_R^2 E(J))/(2(20R+41))$ for every $J \geq R$.

Outline of the formal proof. At $M = 2R+2$ the constants are $88R+182 = 2(11(2M+4)+3)$, $20R+41 = 5(2M+4)+1$ and $f_R = 5(2M+4)(M+1)$. Each jump is at most L, so $E(J_0) \leq 1$.

(a) Either $\Gamma_{J_0}(j) = 0$, or it is the jump of an actual landing step (u, F) at a selection of the first $J' \leq J_0$ families. Then F also belongs to the first $J \geq J_0$ families, and Lemma 5.6(b) at the selection $(J, R, 2R+2)$ and any of its legal phases gives $\mathcal{A}(S_\rho(s_j)) \geq (\Gamma_{J_0}(j)/L)^2/(88R+182)$; the family need not survive there. Average over legal phases and columns.

(b) For $J \geq R$, the selection $(J, R, 2R+2)$ is among those that define $\Gamma_J(j)$, so $\Gamma_\rho(s_j) \leq \Gamma_J(j)$ at each of its legal phases. With $y = \Gamma_J(j)/L$ and $\alpha = f_R \Gamma_\rho(s_j)/L \leq f_R y$,

$$(1-\alpha)_+^2 \geq 1 - 2\alpha \geq 1 - 2f_R y \geq \tfrac{1}{2} - 2f_R^2 y^2,$$

the last step because $2(f_R y - \frac{1}{2})^2 \geq 0$. Apply Lemma 5.6(c) and average over phases and columns. □

Theorem 5.8 (Theorem B: single-sweep terminal landings are excluded). Lean:
`SourceCovers.NearFullSequence.not_eventually_terminalLanding`,
`SourceCovers.NearFullSequence.not_frequently_terminalLanding`
[`GapAreaMemo/TerminalLandings.lean`]

In no near-full sequence are all geometries from some index on single-sweep terminal-landing geometries, and in none are infinitely many of its geometries such geometries. The parameters c, β, N, q of the sequence are arbitrary, and no envelope on the sizes of the core families is assumed.

Outline of the formal proof. Fix a joint profile; along its subsequence the geometries eventually carry such data.

Step 1: the landing energies vanish. For every J_0, the landing energy of the geometry of index ν tends to zero along the subsequence. Otherwise it is at least some $\varepsilon > 0$ infinitely often. By Lemma 5.7(a), for every R and every $J \geq \max(J_0, R)$ the inner upper limit of $\overline{\mathcal{A}}_{\nu,J,R,2R+2}$ is at least $\varepsilon/(88R+182)$, so $\mathcal{A}^*_R \geq \varepsilon/(88R+182)$. Theorem G gives $\varepsilon/(88R+182) \leq \beta\tau_R/(R+2)$, and since $88R+182 \leq 91(R+2)$, $\varepsilon \leq 91\beta\tau_R$ for every R, contrary to $\tau_R \to 0$.

Step 2: the selections themselves. Fix R and $J \geq R$. By Step 1, eventually $2f_R^2 E(J) \leq 1/4$, and Lemma 5.7(b) gives $\overline{\mathcal{A}}_{\nu,J,R,2R+2} \geq 1/(8(20R+41))$. Hence $\mathcal{A}^*_R \geq 1/(8(20R+41))$, and Theorem G gives

$$\beta\tau_R \geq \frac{R+2}{8(20R+41)} \geq \frac{1}{164} \qquad (R \geq 0),$$

contrary to $\tau_R \to 0$. If such geometries occur infinitely often, they form a near-full subsequence, and the first case applies to it. □

Why it matters. The two-exception class with an inverse-square hop envelope is the nearest unresolved case of [2, Section 9.1], and the necessary hop-tail bound [2, Theorem 4.2] is compatible with it. Theorem B excludes, with no envelope, the two-exception geometries in which the older family carries the only raw sweep and receives terminal landings. A near-full sequence whose geometries have at most two exceptional families must therefore, from some index on and for every choice of e, g and x_0 with $\mathcal{E} \subseteq \{e, g\}$, violate one of (L2)–(L5): it reads a coordinate other than x_0 during the lower region of a core family or of g, or at a column root; or its raw path decreases somewhere on the lower region of e; or a core family or g acts during that lower region. For an impossibility proof the argument shows how Theorem G is used at two scales: a fixed prefix bounds the landing jumps of every larger selection (Lemma 5.7(a)), and at the selection itself few starts with small jumps confine the frozen raw set to a few short intervals (Lemma 5.7(b)).

6 Sharpness for two exceptional families

Theorems A and B exclude two classes by proving lower bounds on frozen point areas. The source-cover paper excludes every common power envelope $|G_i| \leq HL/i^p$ with $p > q$ [2, Corollary 4.4]; for two exceptional families the inverse-square envelope $p = 2$ is the boundary case. This section shows that the frozen gap-area condition (GA) cannot exclude that boundary case at moduli comparable to R. We construct admissible geometries with two exceptional families and hop counts $|G_i| \leq 512L/i^2$ whose frozen point areas have mean at most $10^7/R^2$ over legal phases and columns, at every modulus, and whose hop tails are at least $1/(32R)$. Hence (GA) holds on them whenever $3.2\cdot 10^8(M-R) \leq \beta R$, for instance at $M = 2R+2$ for every $\beta \geq 10^9$ (Theorem 6.6 and Corollary 6.7). The geometries carry a single input. Frozen point areas and hop profiles do not involve the correctness sets, so this is a statement about what the certificate can see; it constructs no transposer.

The idea. Two banks of zigzags, A and B, carry the root of every column down through bank A, across the newer exceptional family g into bank B, and down through bank B; the input head does not move during them. Bank A turns the column and the depth p at which the chain leaves it into an offset o, one of about 2^ℓ values when the protected zigzags have levels up to ℓ. Bank B carries every such offset to each of its zigzags Z^{B}_k. A third bank, C, is swept once by the input head. The older exceptional family e sends the descent and the first rise of Z^{B}_k into the second rise of the bank-C zigzag Z^{C}_k, at relative levels $\mathrm{vdC}(k) \mp o/4$ around a centre given by the base-2 van der Corput sequence. The landing levels of one column, about 4^ℓ in number, fill the unit interval up to $O(4^{-\ell})$ (Lemma 6.4); a single bank would give only about 2^ℓ of them. The chains of bank C then

copy every landing level into every earlier zigzag of bank C, so the frozen raw values fill the input range except for one arc of length $O(L/2^\ell)$ (Lemma 6.5). Landings that fall outside a second rise are parked elsewhere and only add points.

Definition 6.1 (The dyadic sharpness geometries)**.** Lean: `SourceCovers.Z3D.Index.geometry`, `SourceCovers.Z3D.Index.runtime_le` [`GapAreaMemo/Z3DGeometry.lean`]

Fix an integer $\nu \geq 3$ and put $d = 2^{2^\nu}$, so that $b = b(d) = 2^\nu$ and $L = d^2 b = 2^{2^{\nu+1}+\nu}$. Each bank has zigzags with indices $1 \leq k \leq k_\nu = 2^{\nu+1} - 1$; the index k has level $\ell(k) = \lfloor \log_2 k \rfloor \leq \nu$ and weight $w_k = 4^{-\ell(k)-1}$. Banks A and B have heights $w_k L$ and bank C has heights $8w_k L$; consecutive heights differ by a factor 1 or 4. A zigzag of height h rises h cells, descends h cells and rises h cells again; every leg moves one cell per step, except that one rising leg may be slowed by a constant factor. Write vdC for the base-2 van der Corput sequence, which reverses binary digits: $\mathrm{vdC}(\sum_i k_i 2^i) = \sum_i k_i 2^{-i-1}$ for digits $k_i \in \{0, 1\}$.

(i) *Bank C* consists of $Z_1^{\mathrm{C}}, \ldots, Z_{k_\nu}^{\mathrm{C}}$ in this time order. Z_1^{C} starts at time 1; for $k \geq 2$, the first rise of Z_k^{C} runs during the second rise of Z_{k-1}^{C}, slowed by the factor w_{k-1}/w_k. Let U_k be the trough of Z_k^{C}, and let $U^* = U_{k_\nu} + 8w_{k_\nu} L$ be the end of bank C.

(ii) *Bank B* consists of $Z_{k_\nu}^{\mathrm{B}}, \ldots, Z_1^{\mathrm{B}}$ in this time order, after bank C. The second rise of Z_k^{B} runs during the first rise of Z_{k-1}^{B}, slowed by the factor w_{k-1}/w_k; that of Z_1^{B} is slowed by 4 and takes L steps. Bank B ends at the trough O_B of Z_1^{B}.

(iii) *Bank A* consists of zigzags Z_k^{A} laid out like those of bank B, after the second rise of Z_1^{B}; it ends at the trough O_A of Z_1^{A}. The outputs are appended during the second rise of Z_1^{A}, at $t_z = O_A + 1 + z$ for $z < L$.

(iv) *Heads.* One work head performs the zigzags of odd index of the three banks in time order, another those of even index. Each zigzag starts at a fresh cell above all cells its head has visited before. After the halting time each of these heads performs padding zigzags; complete families are formed from hops before the halting time only, so the padding adds no family. Two further work heads perform one excursion each. The newer family g rises at unit speed from $O_B + 1$ and descends at unit speed during bank A, reaching level i at time $O_A - i$, for $i < h_g = 2\sum_k w_k L$. The older family e descends at unit speed during bank B, reaching level l at time $O_B - l$, and rises earlier. Level 0 departs at time 0; for $l \geq 1$, the level that reaches the bank-B time at offset $0 \leq r < 2w_k L$ from the start of Z_k^{B} departs at the landing time

$$U_k + \psi_k\big(\lfloor 4w_k L\,\mathrm{vdC}(k) \rfloor + w_k L - r\big),$$

where $\psi_k(z) = 2z$ for $0 \leq z \leq 4w_k L$ (a landing in the second rise of Z_k^{C}), $\psi_k(z) = z$ for $z < 0$ and $\psi_k(z) = 4w_k L + z$ for $z > 4w_k L$ (parking in the descents of Z_k^{C} and Z_{k+1}^{C}, or, for $k = k_\nu$, shortly after bank C).

(v) *Input head.* It sweeps once across bank C: $x(t) = \min(\lfloor (t-1)/\sigma \rfloor, L-1)$ with $\sigma = \lfloor (U^* - 1)/L \rfloor$.

(vi) *Computation.* A single input and a one-point record alphabet.

The original core consists of the upper and lower families of the $3k_\nu$ bank zigzags; the exceptional families are e and g; and the bound on all other families is $a_0 = 0$. We write Γ_ν for this geometry.

The Lean namespace is `Z3D`, for three banks of zigzags with dyadic parameters. The parameters are `SourceCovers.Z3D.Index`, the heads `SourceCovers.Z3D.Index.heads`, the raw path and the append times `SourceCovers.Z3D.Index.raw` and `SourceCovers.Z3D.Index.out`, the two exceptional schedules `SourceCovers.Z3D.Index.eSchedule` and `SourceCovers.Z3D.Index.gSchedule`,

and the core `SourceCovers.Z3D.Index.core`. Each zigzag splices two scheduled excursions of the shared library.

Lemma 6.2 (Admissibility)**.** Lean: `SourceCovers.Z3D.Index.family_mem`, `SourceCovers.Z3D.Index.core_disjoint`, `SourceCovers.Z3D.Index.exceptions_card`, `SourceCovers.Z3D.Index.outside_none` [`GapAreaMemo/Z3DFamilies.lean`]

For every $\beta \geq 0$, Γ_ν is an admissible geometry with $c = 128$, $N = 3$, $q = 2$ and $a_0 = 0$. Every complete family is a core family, e or g; the core has a two-class span partition and the full family system a three-class one.

Outline of the formal proof. The walk of each of the first two heads is a sequence of zigzags at fresh cells, and its complete families are exactly the upper families of its zigzags (the returns of the first rise and the descent, owned at the peak) and the lower ones (the returns of the descent and the second rise, owned at the trough); each exceptional excursion is one native family. Hence every family is a bank family or an exception. Upper families of distinct bank zigzags have disjoint spans, and so do lower families; colouring the core by upper and lower gives its two classes (`SourceCovers.Z3D.Index.core_coloring`). The spans of e and g are disjoint, and they form the third class (`SourceCovers.Z3D.Index.full_coloring`). The banks and the output phase take at most $128L$ steps. □

Lemma 6.3 (Inverse-square sizes and protection)**.** Lean: `SourceCovers.Z3D.Index.famA_card`, `SourceCovers.Z3D.Index.famB_card`, `SourceCovers.Z3D.Index.famC_card`, `SourceCovers.Z3D.Index.card_big_le`, `SourceCovers.Z3D.Index.hop_envelope`, `SourceCovers.Z3D.Index.famA_protected`, `SourceCovers.Z3D.Index.famB_protected`, `SourceCovers.Z3D.Index.famC_protected`, `SourceCovers.SortedCore.prefix_retained` [`GapAreaMemo/Z3DProtected.lean`]

In Γ_ν, the upper and lower families of Z_k^{A} and of Z_k^{B} have $w_k L$ hops each, and those of Z_k^{C} have $8w_k L$ hops. For every $\ell \geq 0$, at most $16 \cdot 2^\ell - 6$ core families have at least $L\,4^{-\ell-1}$ hops. Hence, in every order $G_1, G_2, \ldots$ of the core by nonincreasing hop count:

(a) $|G_i| \leq 512L/i^2$ for every i;

(b) if $16 \cdot 2^\ell \leq R$, the first R families contain the upper and lower families of every Z_k^{A} and Z_k^{B} with $\ell(k) \leq \ell$ and of every Z_k^{C} with $\ell(k) \leq \ell + 1$; and for all $J \geq R$ and M, every family among the first R survives at every legal phase.

Outline of the formal proof. Each core family is the upper or lower family of one bank zigzag, and its hops are the returns of one pair of legs, one per cell. The families with at least $L4^{-\ell-1}$ hops belong to zigzags Z_k^{A}, Z_k^{B} with $\ell(k) \leq \ell$ and Z_k^{C} with $\ell(k) \leq \ell + 1$, at most $4(2^{\ell+1} - 1) + 2(2^{\ell+2} - 1)$ families. (a) If $L4^{-\ell-1} \leq |G_i| < L4^{-\ell}$, then $G_1, \ldots, G_i$ all have at least $L4^{-\ell-1}$ hops, so $i < 16 \cdot 2^\ell$; and every family has at most $2L$ hops. (b) If a family with at least $L4^{-\ell-1}$ hops stood at position $p > R$, the p distinct families up to it would all have that many hops, more than $16 \cdot 2^\ell$. A protected family survives at a legal phase [2, Lemma 2.6]. □

Lemma 6.4 (Van der Corput rows with dyadic offsets)**.** Lean: `SourceCovers.Covering.vdC`, `SourceCovers.Covering.vdC_hits`, `SourceCovers.Covering.w`, `SourceCovers.Covering.t`, `SourceCovers.Covering.t_sub_succ`, `SourceCovers.Covering.offsets`, `SourceCovers.Covering.offsets_dense`, `SourceCovers.Covering.row_cover_quarter`, `SourceCovers.Covering.rowPoints_dense_of_le` [`GapAreaMemo/Covering.lean`]

For $p \geq 1$ let w_p be as in Definition 6.1, put $t_p = (2^{\ell(p)+1} - p)2^{-2\ell(p)-1} + 2^{-\ell(p)-1}$, so that $t_1 = 1$ and $t_p - t_{p+1} = 2w_p$, and for $D \geq 1$ and rational $\theta \in [0, 1]$ let $O_D(\theta) = \{t_p - \theta w_p,\ t_p - (2 - \theta)w_p : 1 \leq p \leq D\}$. Let $n \geq 6$ and $2^n \leq 16D$. Then for every $k_0 \geq 0$ the points $\mathrm{vdC}(k) \pm o/4$ with

$k_0 \le k < k_0 + 2^n$ *and* $o \in O_D(\theta)$ *meet every interval* $[x, y] \subseteq [0, 1]$ *with rational endpoints and* $y - x \ge 1152 \cdot 4^{-n}$.

Outline of the formal proof. Any 2^n consecutive values of vdC meet every dyadic interval of length 2^{-n}, hence every interval of length 2^{1-n} in $[0, 1]$. The offsets lie in $(0, 1]$, and above every $r \in [t_D, 1/2]$ some offset lies in $[r, r + 8r^2]$, because consecutive values t_p differ by $2w_p$ and $w_p \le r^2$ when $t_{p+1} \le r$. Given $[x, y]$, choose a centre $\mathrm{vdC}(k)$ at a distance of order 2^{-n} from one end of the interval and an offset close to four times that distance: the row point $\mathrm{vdC}(k) \pm o/4$ then lies in $[x, y]$. These covering properties are elementary, and we claim no novelty for them. □

Lemma 6.5 (Point area from dense values)**.** Lean: `SourceCovers.GapBound.window_cover`, `SourceCovers.GapBound.pointArea_le_of_dense'` [`GapAreaMemo/GapBound.lean`]

(a) Let $\alpha_0, \dots, \alpha_n$ *and* $w'_0, \dots, w'_{n-1}$ *be natural numbers with* $\alpha_s + w'_s \le \alpha_{s+1}$, *and let* $\Lambda \ge 1$ *be at least every* w'_s *and every* $w'_s + w'_{s+1}$. *If every window* $[z, z + w'_s] \subseteq [\alpha_s, \alpha_{s+1}]$ *meets a set* S *of natural numbers, then every window* $[z, z + \Lambda] \subseteq [\alpha_0, \alpha_n]$ *meets* S.

(b) Let $S \subseteq \Omega$ *be nonempty with* $L - 1 \in S$, *and let* $\gamma \le X \le L$ *be integers such that every window* $[u, u + \gamma] \subseteq [0, X]$ *meets* S. *Then*

$$\mathcal{A}(S) \le \frac{(\gamma + 1)L + (L - X + \gamma + 1)^2}{2L^2}.$$

Outline of the formal proof. (a) Let $\alpha_s \le z < \alpha_{s+1}$. If $z + w'_s \le \alpha_{s+1}$, use the window $[z, z + w'_s]$; otherwise use $[\alpha_{s+1}, \alpha_{s+1} + w'_{s+1}]$, which lies in $[z, z + \Lambda]$. (b) A gap starting at a point u with $u + \gamma + 1 \le X$ has length at most $\gamma + 1$. The other gaps that do not wrap around lie in $[X - \gamma, L - 1]$ and have total length at most $L - X + \gamma + 1$, and the wraparound gap ends at a point at most γ. Bound the sum of the squared gaps by the largest gap of each kind times the total length of that kind. □

Theorem 6.6 (Theorem Z: frozen gap areas of the sharpness geometries)**.** Lean: `SourceCovers.Z3D.Index.phase_le`, `SourceCovers.Z3D.Index.mean_le`, `SourceCovers.Z3D.Index.gap_area_le`, `SourceCovers.Z3D.Index.hop_tail`, `SourceCovers.Z3D.Index.gap_area_condition`, `SourceCovers.Z3D.Index.gap_area_condition_default` [`GapAreaMemo/Z3DSharp.lean`]

Let $\nu \ge 3$ *and* $\beta \ge 0$, *and fix an order* $G_1, G_2, \dots$ *of the core of* Γ_ν *by nonincreasing hop count.*

(a) If $1 \le R \le 2^\nu$, $M \ge R + 1$ *and* $J \ge R$, *then* $\overline{\mathcal{A}}_{J,R,M} \le 10^7/R^2$.

(b) If $R \ge 1$, $4R \le 2^\nu$ *and* $J \ge 64R$, *then* $\sum_{R<i\le J} |G_i| \ge L/(32R)$.

(c) Hence, if $R \ge 1$, $4R \le 2^\nu$, $J \ge 64R$, $M \ge R + 1$ *and* $3.2 \cdot 10^8(M - R) \le \beta R$, *then*

$$\overline{\mathcal{A}}_{J,R,M} \le \frac{\beta}{M - R} \sum_{R<i\le J} \frac{|G_i|}{L}.$$

In particular this holds for $M = 2R + 2$ *whenever* $\beta \ge 10^9$.

Outline of the formal proof. (a) For $R^2 \le 2 \cdot 10^7$ the bound is at least $1/2$. Otherwise choose $\ell \ge 8$ with $16 \cdot 2^\ell \le R < 32 \cdot 2^\ell$. By Lemma 6.3(b), at every legal phase the orbit families contain the families of all bank zigzags of indices below $2^{\ell+1}$ and both exceptions. Fix a column $j \ge 1$ and put $\theta = j/d$.

(i) *Bank A.* The root s_j lies in the second rise of Z_1^{A} at level $\theta w_1 L$. The lower family of Z_1^{A} maps it into the descent, the upper family into the first rise, which runs during the second rise of Z_2^{A}, and so on: Z_p^{A} is reached at the exact level $\theta w_p L$, in its descent and in its first rise (`SourceCovers.Z3D.Index.late_chain`).

(ii) *Across g.* These times lie in the descent of g, which maps them into the second rise of Z_1^{B} at relative level $1-o$, with $o = t_p - \theta w_p$ from the descent and $o = t_p - (2-\theta)w_p$ from the first rise, because $2\sum_{i<p} w_i = 1 - t_p$.

(iii) *Bank B and e.* The chain of bank B reaches Z_k^{B} at a level within 3 of $(1-o)w_k L$, the levels being rounded down. The family e maps the descent time of Z_k^{B} into the second rise of Z_k^{C} at a height 2μ whose relative level $(2\mu+1)/(8w_k L)$ lies within $1/(w_k L)$ of $\mathrm{vdC}(k) - o/4$, and the first-rise time to one within $1/(w_k L)$ of $\mathrm{vdC}(k) + o/4$ (`SourceCovers.Z3D.Index.landing`).

(iv) *Density.* By Lemma 6.4 with $k_0 = 2^\ell$, $n = \ell$ and $D = 2^\ell$, these relative levels, for $2^\ell \le k < 2^{\ell+1}$, meet every window inside $[2/(w_k L), 1 - 2/(w_k L)]$ of relative length about $1152 \cdot 4^{-\ell}$ (`SourceCovers.Z3D.Index.levels_window`).

(v) *Bank C.* The chain of bank C carries a landing at height η in Z_k^{C} to every Z_i^{C} with $i \le k$, at the height η' with $(\eta'+1)/(8w_i L) = (\eta+1)/(8w_k L)$, in its second rise, its descent and its first rise (`SourceCovers.Z3D.Index.early_chain`). Hence every window of relative length about $1152 \cdot 4^{-\ell}$ inside the first rise or the descent of each of $Z_1^{\mathrm{C}}, \dots, Z_{2^\ell}^{\mathrm{C}}$ meets the orbit, and by Lemma 6.5(a) so does every time window of length $4608L4^{-\ell} + O(4^{\ell})$ that ends by the end of the first rise of $Z_{2^\ell}^{\mathrm{C}}$ (`SourceCovers.Z3D.Index.time_cover`).

(vi) *Raw values.* Since x advances one cell every $\sigma \ge 1$ steps, the frozen raw set meets every window of that length inside $[0, X]$. Here $L - X \le 40L/2^\ell + O(L4^{-\ell})$, because the rest of bank C lasts at most $40\sum_{k>2^\ell} w_k L + 16 w_{2^\ell} L$ steps and $\sigma \ge 2$; and $L - 1 = x(s_j)$ lies in the set. Lemma 6.5(b) gives $\mathcal{A}(S_\rho(s_j)) \le 4010 \cdot 4^{-\ell}$ (`SourceCovers.Z3D.Index.phase_bound`).

None of this depends on M. Averaging over legal phases and columns, with the column $j = 0$ bounded by $1/2$, gives at most $4011 \cdot 4^{-\ell} < 4.2 \cdot 10^6/R^2$. (b) Let 2^r be the power of two with $2R < 2^r \le 4R$. The upper families of the zigzags Z_k^{A} with $2^r \le k < 2^{r+1}$ are 2^r distinct families of $L4^{-r-1}$ hops each. Fewer than $16 \cdot 2^r \le J$ core families have that many hops, so all of them lie among the first J (Lemma 6.3); at most R of them lie among the first R; hence $\sum_{R<i\le J} |G_i| \ge (2^r - R)L4^{-r-1} \ge L/(32R)$. (c) Combine (a) and (b): $10^7/R^2 \le \beta/(32R(M-R))$; for $M = 2R+2$ use $M - R \le 3R$. □

Corollary 6.7 (Limit form)**.** Lean: `SourceCovers.Z3D.seq`, `SourceCovers.Z3D.limit_form`, `SourceCovers.Z3D.limit_GA`, `SourceCovers.Z3D.limit_GA_zero`, `SourceCovers.Z3D.limit_GA_default` [`GapAreaMemo/Z3DLimit.lean`]

Consider the sequence $(\Gamma_{\nu+3})_{\nu\ge 0}$ with any orders of its cores by nonincreasing hop count, and any strictly increasing subsequence along which the normalized hop profiles $|G_i|/L$ converge to limits h_i. Define $\mathcal{A}^*_{R,M}$ and $\tau_R = \sum_{i>R} h_i$ as in Definition 3.1 and Section 2. Then $\mathcal{A}^*_{R,M} \le 10^7/R^2$ and $\tau_R \ge 1/(32R)$ for all $R \ge 1$ and $M \ge R+1$. Hence

$$\mathcal{A}^*_{R,M} \le \frac{\beta \tau_R}{M - R}$$

whenever $R \ge 1$, $M \ge R+1$ and $3.2 \cdot 10^8 (M-R) \le \beta R$, and whenever $R = 0$ and $2 \le M \le \beta/16$. In particular (GA) holds at $M = 2R+2$ for every $R \ge 0$ when $\beta \ge 10^9$.

Outline of the formal proof. For fixed $J \geq R$, the geometries of the subsequence eventually satisfy $R \leq 2^{\nu}$, and Theorem 6.6(a) bounds the inner upper limit; the outer one preserves the bound. The finite profile masses are at most 128, so the limit profile is nonnegative and summable; Theorem 6.6(b) with $J = 64R$ passes to the limit, and $\tau_R \geq \sum_{R<i\leq 64R} h_i$. For $R = 0$, $\mathcal{A}^*_{0,M} \leq 1/2$ and $\tau_0 \geq \tau_1 \geq 1/32$. □

Why it matters. For an impossibility proof, Theorem Z closes one route: the frozen gap-area condition at moduli up to $R + \beta R/(3.2 \cdot 10^8)$ cannot exclude the two-exception inverse-square class. An exclusion of that class must use larger moduli, the remaining necessary conditions of [2], or input specific to transposition (Problem 7.2). For a construction, the geometries show how two exceptional families can meet (GA): the older family lands the chains of the protected banks on both sides of well-spread centres in a bank that the input head sweeps, and the chains of that bank merge the landings of different zigzags.

The sequence of Corollary 6.7 belongs to the class of Problem 7.1 for all $c \geq 128$, $N \geq 3$, $H \geq 512$ and $\beta > 0$, with $a_0 = 0$, and (GA) holds along every joint profile at every modulus allowed in Corollary 6.7. The argument of Remark 3.3 does not use near-fullness; applied to this sequence (proof sketch only), it shows that the weighted reciprocal-cover bound [2, (4.8)] holds at the same moduli. By Lemma 4.2(c), Theorem 6.6 also implies that for large ν the geometries have no funnel of a fixed depth over a raw interval of fixed positive relative length; and they are not single-sweep terminal-landing geometries, since the input head moves on the lower regions of the bank-C families. These two observations are elementary and are not formalized. The constants are not optimized. Nothing here concerns larger moduli, the necessary complete-observable movement and global continuation density of [2, Corollaries 4.8 and 5.3], or the existence of a transposer with such geometries.

7 What remains

Scope. Theorems G, A and B are certificates for specified geometries. They assume the admissible hypotheses of [2, Definition 2.2]: one fixed full geometry with a Q_2 core, a bounded number of exceptional families, a bounded full-span partition, and fixed runtime and capacity constants. These hypotheses are not derived for an arbitrary linear-time transposer, and passing from an adaptive machine to one geometry may lose entropy that must be paid for separately [2, Sections 1.1 and 9.3]. Nothing here decides (T).

What a construction in the admissible class must do. For an oblivious transposer that is correct on every input, $\Delta = 0$ on its single geometry [2, Section 1.1]. If its geometries are admissible with fixed parameters and $a_0 = o(L)$, they form a near-full sequence as $d \to \infty$; more generally, every sequence of admissible geometries with $d \to \infty$, $a_0 = o(L)$ and $\Delta = o(L)$ is near-full. Such a sequence must satisfy (GA) at every protected scale, together with the necessary conditions of [2, Theorems 4.2 and 4.6 and Corollaries 4.8 and 5.3], the last two for two exceptional families. By Theorem A, for every fixed depth m and every fixed $\delta > 0$ its family systems eventually admit no funnel of depth m containing all reads of a raw interval of relative length δ. By Theorem B, its geometries eventually admit no single-sweep terminal-landing data. These are necessary conditions, not constructions of a transposer. By Theorem Z, the condition (GA) at moduli $M \leq R + \beta R/(3.2 \cdot 10^8)$ is compatible with two exceptional families and an inverse-square hop envelope.

Open problems.

Problem 7.1 (Frozen gap areas for two exceptions)**.** *Fix $c, \beta > 0$, $N \geq 1$ and $H \geq 0$. Consider sequences of admissible geometries with at most two exceptional families and a common envelope $|G_i| \leq HL/i^2$ in the supplied hop order, with $d \to \infty$ and $a_0/L \to 0$ but no condition on the correctness deficit. Joint profiles and $\mathcal{A}^*_{R,M}$ are defined for them exactly as in Section 2 and Definition 3.1. Does every such sequence have a joint profile with $\mathcal{A}^*_{R,M} > \beta\tau_R/(M-R)$ for some R and M? By Theorem G, a positive answer would exclude every near-full sequence in the class; a negative answer would show that Theorem G alone cannot exclude it. For $c \geq 128$, $N \geq 3$ and $H \geq 512$, the sequence of Corollary 6.7 shows that no pair (R, M) with $R \geq 1$ and $3.2 \cdot 10^8(M-R) \leq \beta R$, and no pair with $R = 0$ and $M \leq \beta/16$, suffices; for $\beta \geq 10^9$ this includes $M = 2R+2$ for every R. The question remains open for larger moduli relative to R and for smaller constants. In particular, for the sharpness geometries themselves it is open whether* (GA) *holds at every R and M, and for which β.*

Problem 7.2 (Transposition-specific input)**.** *The certificates here use the transposition demands only through the cyclic-freezing bound inside Theorem G [2, Theorem 2.9]. The curved-motion theorem of the fourth paper, [1, Theorem 3.12] (Theorem Cq, proved there and not formalized), states that in every macroscopic limit of a linear-time transposer the work heads travel a total distance of at least half a block unit along motion that is not countably-piecewise affine. Can such constraints on the work heads exclude, or bound the frozen point areas of, admissible two-exception geometries that have no funnels of bounded depth and are not single-sweep terminal-landing geometries? The sharpness geometries of Definition 6.1 are a test case. Their work heads move along affine legs, so in the macroscopic limit their paths are countably-piecewise affine. Theorem Cq does not apply to them as it stands: its hypothesis is an inequality that the first paper's Theorem M* proves for every limit of every machine [1, Section 3], while Γ_ν carries a single input and is not the run of a transposer. Bringing constraints of this kind into the admissible-geometry framework is the open step. It is also not established here whether Γ_ν meets the necessary complete-observable movement and global continuation density of [2, Corollaries 4.8 and 5.3].*

Deciding (T) needs more than these certificates: for instance one fixed machine with a proof that it transposes every input in $O(L)$ steps, or, as one possible route to the negative answer, a reduction of every candidate machine to admissible geometries and correctness sets with a positive net deficit, including the cost of selecting a geometry in the adaptive case [2, Section 9.3].

A Index of Lean declarations

The status lines of the memo cite the following declarations, listed by Lean module with their fully qualified names. Module files are relative to the memo's `lean/` directory. The axiom checks in `lean/MemoAxioms.lean` are generated from the same sources (Appendix B).

`GapAreaMemo.CoreWords` `SourceCovers.coreSuccess_internal_nonleaf`, `SourceCovers.coreSuccess_prefix_or`, `SourceCovers.core_root_count`.

`GapAreaMemo.Covering` `SourceCovers.Covering.offsets`, `SourceCovers.Covering.offsets_dense`, `SourceCovers.Covering.rowPoints_dense_of_le`, `SourceCovers.Covering.row_cover_quarter`, `SourceCovers.Covering.t`, `SourceCovers.Covering.t_sub_succ`, `SourceCovers.Covering.vdC`, `SourceCovers.Covering.vdC_hits`, `SourceCovers.Covering.w`.

GapAreaMemo.Funnels SourceCovers.ActualFamily.Feeds,
SourceCovers.AdmissibleGeometry.HasFunnel,
SourceCovers.AdmissibleGeometry.prefixMeanGapArea_ge_of_funnel, SourceCovers.Funnel,
SourceCovers.Funnel.orbit_inter_subset, SourceCovers.Funnel.reach,
SourceCovers.Funnel.reach_card, SourceCovers.NearFullSequence.not_eventually_funnel,
SourceCovers.NearFullSequence.not_frequently_funnel,
SourceCovers.phaseGapArea_ge_of_funnel.

GapAreaMemo.GapArea SourceCovers.AdmissibleGeometry.prefixMeanGapArea,
SourceCovers.AdmissibleGeometry.prefix_gap_area_deficit,
SourceCovers.NearFullSequence.JointProfile.gapAreaLimit,
SourceCovers.NearFullSequence.JointProfile.gapAreaLimit_bounds,
SourceCovers.NearFullSequence.JointProfile.necessary_gap_area,
SourceCovers.NearFullSequence.JointProfile.necessary_gap_area_default,
SourceCovers.NearFullSequence.JointProfile.prefixGapArea,
SourceCovers.NearFullSequence.JointProfile.prefix_gap_area_limit,
SourceCovers.gap_area_paid_deficit, SourceCovers.meanGapArea, SourceCovers.phaseGapArea,
SourceCovers.phaseGapArea_bounds.

GapAreaMemo.GapBound SourceCovers.GapBound.pointArea_le_of_dense',
SourceCovers.GapBound.window_cover.

GapAreaMemo.LandingCounts SourceCovers.IsLandingStep, SourceCovers.landingSet,
SourceCovers.landingStarts, SourceCovers.landingStarts_card, SourceCovers.landing_cover,
SourceCovers.newerClosure, SourceCovers.newerClosure_count,
SourceCovers.newerClosure_count_span, SourceCovers.newerRoots, SourceCovers.stepJump,
SourceCovers.stepMax.

GapAreaMemo.LandingGaps SourceCovers.AdmissibleGeometry.TerminalLanding,
SourceCovers.AdmissibleGeometry.TerminalLanding.gap_of_starts,
SourceCovers.AdmissibleGeometry.TerminalLanding.gap_of_step,
SourceCovers.AdmissibleGeometry.TerminalLanding.raw_values.

GapAreaMemo.LandingOrbits SourceCovers.raw_landing_antitone,
SourceCovers.shadowOrbit_insert_newer, SourceCovers.shadowOrbit_insert_terminal.

GapAreaMemo.TerminalLandings SourceCovers.AdmissibleGeometry.landingEnergy,
SourceCovers.AdmissibleGeometry.landingEnergy_bounds,
SourceCovers.AdmissibleGeometry.landingGamma,
SourceCovers.AdmissibleGeometry.prefixMeanGapArea_ge_later,
SourceCovers.AdmissibleGeometry.prefixMeanGapArea_ge_own,
SourceCovers.NearFullSequence.not_eventually_terminalLanding,
SourceCovers.NearFullSequence.not_frequently_terminalLanding.

GapAreaMemo.Z3DFamilies SourceCovers.Z3D.Index.core_disjoint,
SourceCovers.Z3D.Index.exceptions_card, SourceCovers.Z3D.Index.family_mem,
SourceCovers.Z3D.Index.outside_none.

GapAreaMemo.Z3DGeometry SourceCovers.Z3D.Index.geometry,
SourceCovers.Z3D.Index.runtime_le.

GapAreaMemo.Z3DLimit SourceCovers.Z3D.limit_GA, SourceCovers.Z3D.limit_GA_default, SourceCovers.Z3D.limit_GA_zero, SourceCovers.Z3D.limit_form, SourceCovers.Z3D.seq.

GapAreaMemo.Z3DProtected SourceCovers.SortedCore.prefix_retained, SourceCovers.Z3D.Index.card_big_le, SourceCovers.Z3D.Index.famA_card, SourceCovers.Z3D.Index.famA_protected, SourceCovers.Z3D.Index.famB_card, SourceCovers.Z3D.Index.famB_protected, SourceCovers.Z3D.Index.famC_card, SourceCovers.Z3D.Index.famC_protected, SourceCovers.Z3D.Index.hop_envelope.

GapAreaMemo.Z3DSharp SourceCovers.Z3D.Index.gap_area_condition, SourceCovers.Z3D.Index.gap_area_condition_default, SourceCovers.Z3D.Index.gap_area_le, SourceCovers.Z3D.Index.hop_tail, SourceCovers.Z3D.Index.mean_le, SourceCovers.Z3D.Index.phase_le.

Also named in the text From this memo's library: SourceCovers.Z3D.Index, SourceCovers.Z3D.Index.core, SourceCovers.Z3D.Index.core_coloring, SourceCovers.Z3D.Index.eSchedule, SourceCovers.Z3D.Index.early_chain, SourceCovers.Z3D.Index.full_coloring, SourceCovers.Z3D.Index.gSchedule, SourceCovers.Z3D.Index.heads, SourceCovers.Z3D.Index.landing, SourceCovers.Z3D.Index.late_chain, SourceCovers.Z3D.Index.levels_window, SourceCovers.Z3D.Index.out, SourceCovers.Z3D.Index.phase_bound, SourceCovers.Z3D.Index.raw, SourceCovers.Z3D.Index.time_cover, SourceCovers.pointArea_ge_of_sparse.
From the source-cover paper's library: SourceCovers.NearFullSequence.JointProfile.prefix_error_limit, SourceCovers.core_unique_after, SourceCovers.fractional_phase_gap, SourceCovers.paid_deficit_of_gap_credit.
From the shared library: LeanMachine.IntervalGapArea.point_area_lower_of_budgets.

B Building and checking the proofs

The memo's sources and PDF are licensed under CC BY 4.0, and its Lean library under Apache 2.0. Readers need the whole repository, with sibling directories in place. The memo's Lean package in `memo-gap-area/lean/` requires the source-cover paper's package through the relative path `../../lean`. That package brings in its own dependencies: the second paper's build adapter for the first paper's canonical sources, the eighth paper's `boundedNesting` package and the shared `lean-machine` library, and through them the second and fourth papers' libraries (`transposeFollowup`, `motionDepthCodes`) and the second paper's pinned external dependency, all at the repository's Lean and Mathlib v4.34.1 pins. Fetch or verify that external dependency as described in the second paper's README. No Lean source of an earlier paper is copied or modified; `paper/macros.tex` is an unchanged copy of the source-cover paper's macros.

From the memo's `lean/` directory, run

```
lake exe cache get
lake build
lake env lean MemoAxioms.lean
```

The cache command is needed only when the pinned dependency objects are not already available. The cited declarations depend at most on `propext`, `Classical.choice` and `Quot.sound`. The memo's library uses no `sorry`, no `native_decide`, no new axiom and no heartbeat override.

From the memo directory, regenerate or check Appendix A and `lean/MemoAxioms.lean` with

```
python3 tools/lean_index.py
python3 tools/lean_index.py --check
```

The script reads the status lines and the `\lean` references of the manuscript. It also checks that every numbered result has one status line followed by the matching proof environment, that every declaration in a status line is declared in the cited file, and that every declaration named in the text exists in one of the libraries. It does not run Lean or check proofs.

Build the PDF from `paper/` with

```
latexmk -pdf -interaction=nonstopmode main.tex
```

References

[1] Hans Brende and Claude Opus 5.5 (Anthropic). Curved motion, depth reduction, and storage codes for matrix transposition on tapes. Manuscript and accompanying Lean formalizations, repository HansBrende/papers, October 2026. Pinned revision 5cf90b9e346fee3bb5d7fa233f5b1f1dfec5f447; proof status is specified result by result. URL: https://github.com/HansBrende/papers/tree/5cf90b9e346fee3bb5d7fa233f5b1f1dfec5f447/Curved-Motion-Depth-Reduction-and-Storage-Codes-for-Matrix-Transposition-on-Tapes.

[2] Hans Brende and GPT-6 Astra (OpenAI). Source covers and barriers to tape transposition. Manuscript and accompanying Lean formalizations, repository HansBrende/papers, October 2026. Pinned revision 10b94428cdc199369958661b3fae21096b40aca1; proof status is specified result by result. URL: https://github.com/HansBrende/papers/tree/10b94428cdc199369958661b3fae21096b40aca1/Source-Covers-and-Barriers-to-Tape-Transposition.